CDIS – Y25

Department of CSE

B.Tech. CSE Trimester - 5

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Cloud Infrastructure and Services
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LAB RECORD

2026

Skill Experiment - 3

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AIM:

To launch an Ubuntu EC2 instance, install the Nginx web server automatically using a user data script during instance launch, and access the default Nginx page through a web browser.

Description:

This experiment demonstrates the use of Amazon EC2 to deploy a basic web server on an Ubuntu virtual machine. An Ubuntu Amazon Machine Image (AMI) is selected while launching the instance. A user data script is provided in the advanced launch settings so that the required package installation and service startup commands run automatically when the instance boots. After configuring the security group to allow HTTP traffic, the instance's public IPv4 address is opened in a browser. The supplied output screenshot displays the default “Welcome to nginx!” page, confirming that the web server is running and reachable.

Steps:

Part A: Launch an Ubuntu EC2 Instance

1. Sign in to the AWS Management Console and search for **EC2**.
2. Open the EC2 dashboard and select **Launch instance**.
3. Enter a suitable name for the instance.
4. Under Application and OS Images (Amazon Machine Image), choose an **Ubuntu Server** AMI.
5. Select an instance type that is suitable for the lab exercise.
6. Create or select a key pair if required, and configure the network settings.
7. In the security group rules, allow inbound **HTTP traffic on port 80** from the required source so that the Nginx page can be accessed through a browser.
8. Expand **Advanced details** and locate the **User data** field.

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Steps (Continued):

Part B: Install and Start Nginx Using User Data

9. Enter the following shell script in the User data field. It updates the Ubuntu package list, installs Nginx, enables the service at startup, and starts the web server.

```
#!/bin/bash
apt-get update -y
apt-get install -y nginx
systemctl enable nginx
systemctl start nginx
```

10. Review the instance configuration and click **Launch instance**.

11. Wait until the instance is running and its status checks have passed. The user data script runs during the instance's initial boot; allow a short time for package installation and service startup to finish.

12. Select the instance and copy its **Public IPv4 address** from the instance details.

Part C: Verify the Nginx Web Server

13. Open a web browser and enter the public IPv4 address using HTTP, for example: **http://<Public-IPv4-address>**.

14. Verify that the default Nginx welcome page appears. The page should display the heading “Welcome to nginx!” and indicate that the web server is successfully installed and working.

Differences Observed: Ubuntu with Nginx vs. Amazon Linux with Apache

• **Operating system:** This experiment uses Ubuntu, whereas the comparison setup uses Amazon Linux.

• **Package management:** Ubuntu uses **apt** commands (apt-get update and apt-get install); Amazon Linux commonly uses **yum** or **dnf** depending on its version.

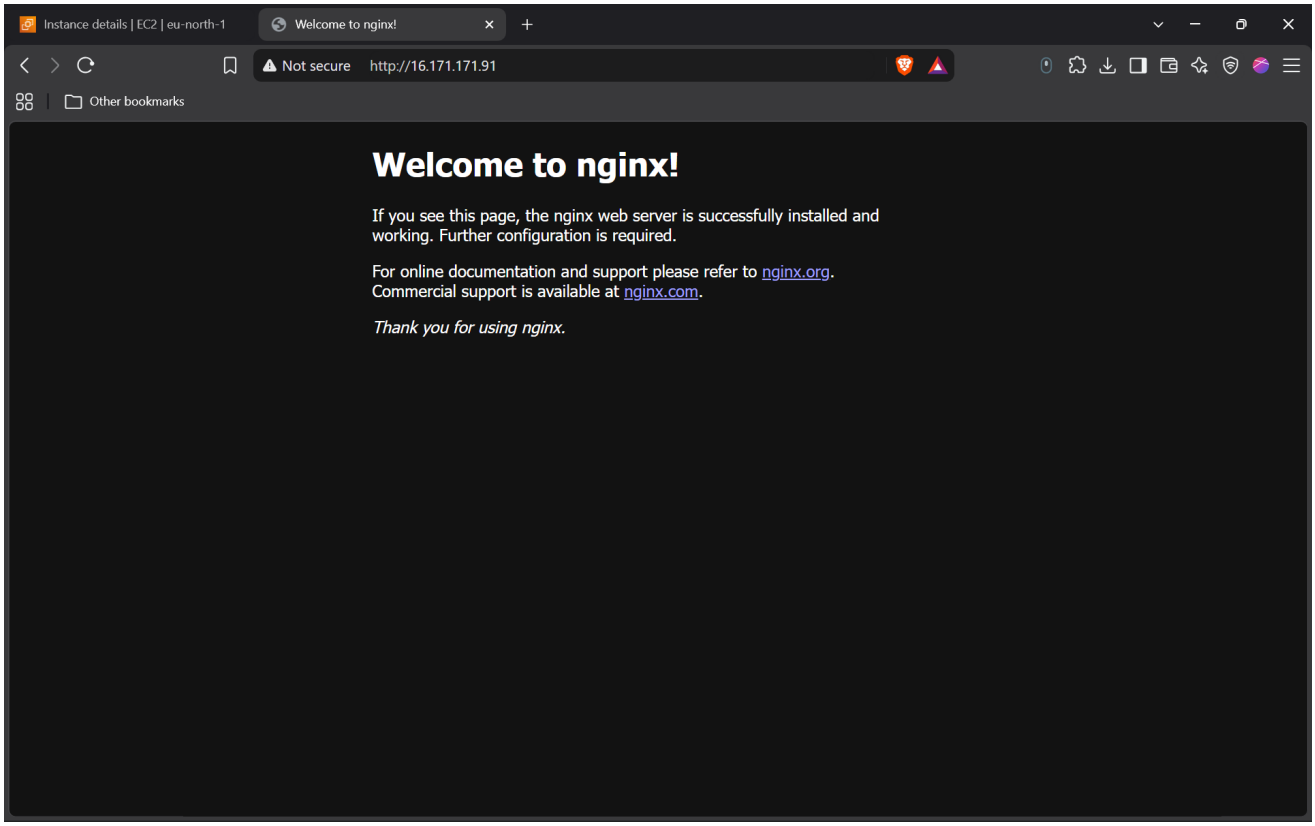
• **Web server:** This experiment installs Nginx, while the comparison setup installs Apache HTTP Server.

• **Service name:** The Nginx service is managed using **systemctl** with the service name **nginx**; Apache on Amazon Linux is commonly managed as **httpd**.

• **Default page:** The browser output in this experiment shows “Welcome to nginx!”, whereas the Apache setup displays its Apache test or default page.

Output:

The public IPv4 address of the Ubuntu EC2 instance was opened in a web browser. The browser displayed the default Nginx welcome page with the message “Welcome to nginx!” and stated that the Nginx web server was successfully installed and working. This confirms that the user data script installed and started Nginx and that HTTP access to the instance was available.



Output Screenshot 1: Ubuntu EC2 instance displaying the default Nginx welcome page in a web browser.

Result: An Ubuntu EC2 instance was launched successfully, and the Nginx web server was installed and started automatically using the user data script. The default Nginx page was accessed through the instance's public IPv4 address, confirming successful deployment.